



TECHNICAL MANUAL

GAME READY[®] THERAPY SYSTEM

Model Numbers:

550450 and 550550-X2 (GR2.0)

550450-X3 and 550550-X3 (GR2.1)

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1. INTRODUCTION

This manual is intended to provide technical information to customers who may benefit from information beyond that provided in the User Manual that accompanies each Game Ready Professional Therapy System. The Technical Manual is intended to be used by individuals who are mechanically inclined and able to follow detailed instructions. It is not intended to REPLACE the User Manual. The User Manuals contains complete information about the Instructions For Use, treatment modes, Warnings, Cautions, Contraindications, and additional information about the use of the system. Use of this manual assumes the user has read and understood the user manual(s) and is familiar with using the Game Ready system. If you do not have a User Manual for your system(s), contact your Game Ready distributor or Game Ready Customer Service at 1.888.426.3732 (from outside the US call +1.510.868.2100) or visit <http://www.gameready.com/support>

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2. THEORY OF OPERATION

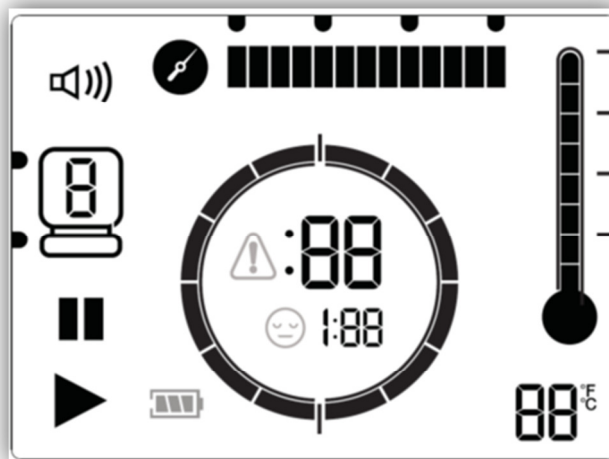
The Game Ready System allows the user to adjust treatment time, temperature and air pressure as desired. Both Manual and “Program Modes” are offered for convenience. The Control Unit comprises a user interface “Control Panel”, a microprocessor, multiple sensors, Hose connection port, and a 12VDC power inlet. The following sections offer additional insight as to how the Control Unit works.

2.1. Self-Diagnostics Checks

When pressing the power button, the Control Unit performs several Self Diagnostic checks.

2.1.1. LCD Display

All segments of the LCD display are cycled on and then off during the self-test. This allows the user to inspect the segments for function.



2.1.2. Air Pressure Sensor “Zero” Check

With the air pump off and the solenoid valve open, the air pressure transducer is checked for an acceptable range (should be close to atmospheric pressure, or zero pressure). If the pressure is beyond acceptable limits, an “01 error” will occur. This could be caused by a bad pressure sensor, or if the Control Unit is hooked up to a Wrap still containing air pressure.

2.1.3. Air Pump Check

If the Air Pressure Sensor “Zero” Check passes, the solenoid valve is closed and the air pump turned on. If the air pressure is not greater than zero within ½ a second, an “02 error” will occur. After this test the air vent is opened and the air pump turned off.

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2.1.4. Fluid Pump Check

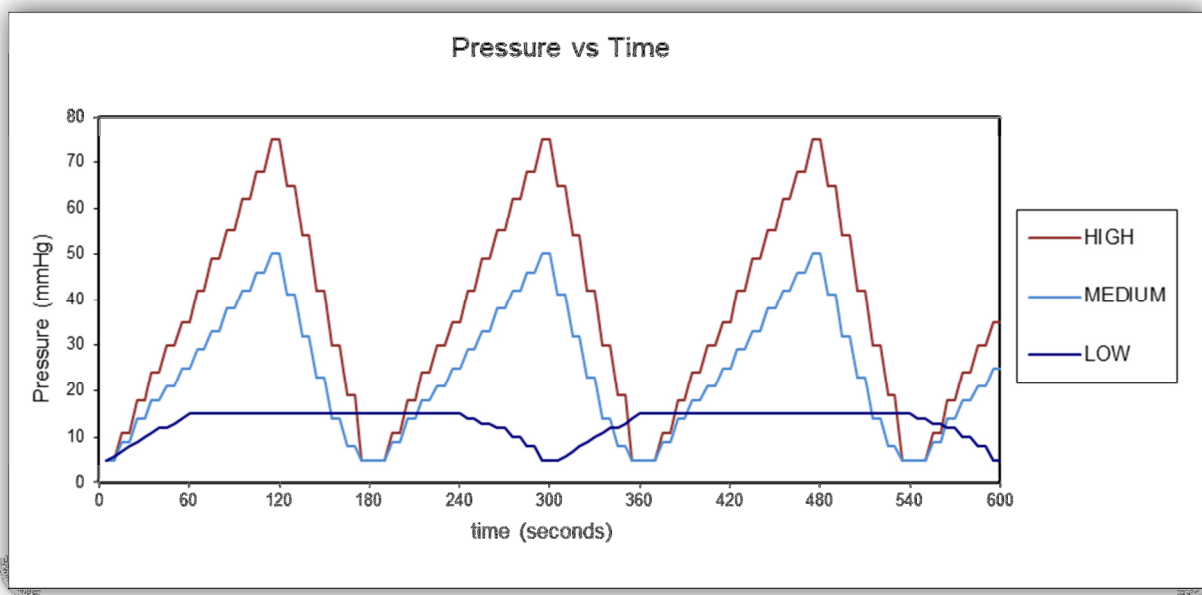
The coolant pump is turned on and monitored for a non-zero current. This is more of a continuity check than a performance check. If no current is sensed, this means that the fluid pump is not working and a “12 error” will occur.

2.1.5. Temperature Sensor Check (GR2.1 only)

The Temperature Sensor is monitored for a “valid value”. Valid values are 28-149 degrees F. It is intended to be a rough check of the sensor, and to prevent operation of the Control Unit in freezing temperatures. If a valid value is not found, a “13 error” will occur.

2.2. Air Pressure

The Control Unit utilizes an air pump, a solenoid vent, and an air pressure sensor that work together to monitor and maintain air pressure in the Wrap according to the pressure setting used. The graph below shows a simplified illustration of the 3 pressure settings.



Air Pressure Indicator

Depending on your software version, the air pressure bar graph will indicate either *actual pressure*, or will indicate the *progress of the pressure cycle*. You can tell which version you have by squeezing a Wrap in the middle of the pressure cycle. If the bar graph is indicating *actual pressure*, it will show an increase from the squeezing of the Wrap.

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Air Pressure Errors

During treatment, the Control Unit continuously monitors and adjusts air pressure. If the air pressure falls outside of certain parameters, air pressure alarms may occur:

- **06:** Over Pressure Error (sensed brief period of over pressure)
- **07:** Under Pressure Error (did not reach top of pressure profile within expected timeframe)
- **08:** Deflation Error (pressure did not vent at the bottom of the pressure profile within the expected timeframe).

2.3. Fluid Flow

The Control Unit has a fluid pump that circulates the ice water through the system. The fluidics circuit contains the ice box filter, a fluid pump, a flow sensing system, and either 1 or 2 temperature sensors (Reference the “Temperature Control” Section for more detail on temperature sensing). The Flow Sensing system monitors your Control Unit for adequate flow returning to the Control Unit.

Fluid Flow Errors: If adequate flow is not sensed, one of two flow alarms will occur. If flow never sensed at the beginning of a treatment, the system assumes the Hose is not connected to a Wrap, there is not enough water in the ice box, there is a blockage in the system, or an air bubble in the fluid pump (pump priming problem). In this case, an 04 error (Dry Pump Alarm) will occur. If flow is sensed at the beginning of a treatment, it will not call an 04 error, but will continue to monitor flow. If flow is blocked during treatment for a long enough period of time, a 10 error (Low Flow) will occur. Also during treatment, the fluid pump current will be monitored. If the pump current is found to be excessive, an 09 error will occur.

2.4. Temperature Control

The Control Unit does not generate cold. The water is cooled by the ice in the ice box, and warmed by water returning from the Wrap, using the heat of the user as a heat source. GR2.0 and GR2.1 systems control temperature in a different manner.

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Temperature Control for GR2.1 Control Units

GR2.1 Control Units utilize a patented temperature control system. The temperature control knob is connected to a digital encoder and allows the user to set the desired temperature (Target Temp.). The Control unit will adjust the fluid pump speed to deliver water from the Control Unit at a temperature close to the Target Temp. This is achieved by using a Baffle inside the GR2.1 ice box. Water returning from the Wrap is typically warmed by the user's body heat. If the water is moving fast (high pump speed), it is sent far into the ice box and readily mixes with the ice and water and cold water is drawn into the baffles. If the water returns more slowly, it falls within the baffle walls and the water temperature inside the baffles becomes warmer than the temperature in the rest of the ice box. Therefore, by varying the pump speed, the temperature of the water leaving the Control Unit may be controlled. This is all done by the Control unit automatically.

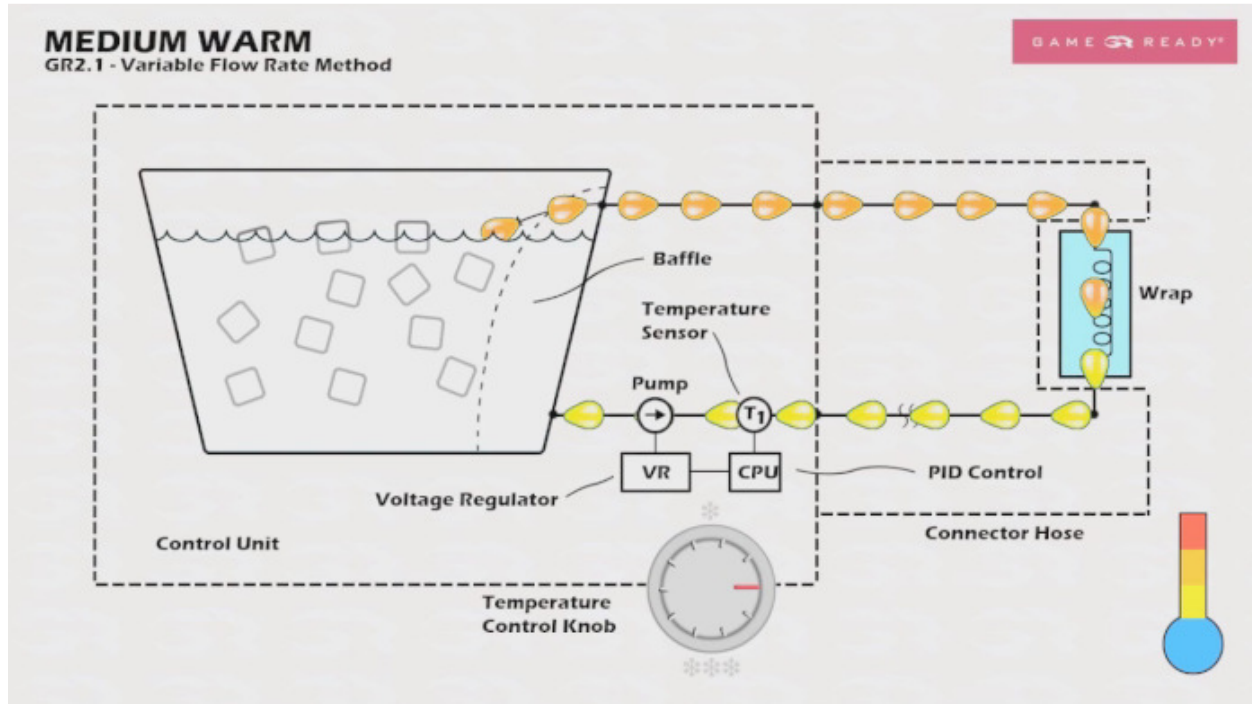
This is a significant improvement over other variable flow rate temperature control systems. With other systems, the water leaving the Control Unit is still ice cold, and in order to achieve a desired "Average Wrap Temperature", much lower flow rates would be required, which leads to significant temperature variation within the Wrap which is undesirable.

Temperature of the water is measured at only one place in the GR 2.1 Control Units - just before leaving the Control Unit (T1). Two different temperatures are displayed on the Control Panel. The "Target Temperature" is the desired temperature, or temperature "set" by the user. The "Actual Temperature" is the temperature of the water leaving the Control Unit (T1).

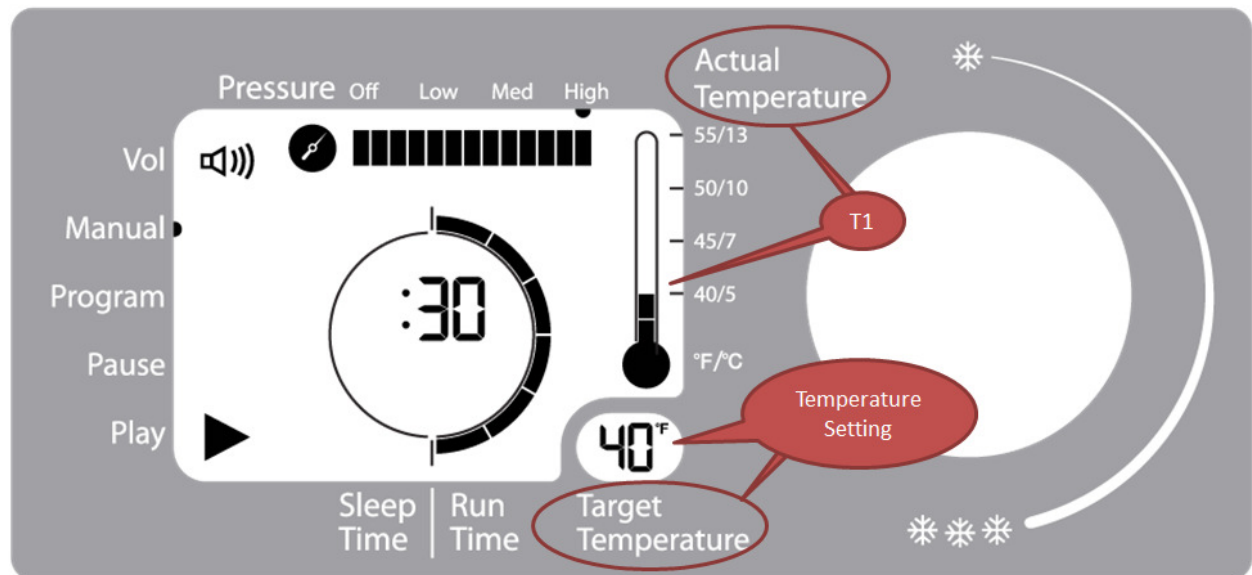
The following notes regarding the GR2.1 Control Unit also may be of interest:

- When set to "full cold" the fluid pump will always pump at full speed regardless of the water temperature.
- If the ice box runs low on ice, there may not be enough ice to achieve desired Target Temperature.
- If the ice box is too full of water (i.e. above the return nozzle) the ability to regulate to warmer temperatures may be limited. Always fill the Control Unit according to the user manual and Fill Labels inside the ice box.
- When the "Play" button is pressed, water will flow at 100% until return flow is sensed by the flow measurement system inside the Control Unit. Once return flow is sensed, the fluid pump will modulate in order to achieve its Target Temperature.
- The maximum set point on a GR2.1 is 50°F (10°C). If warmer temperatures are desired, less or no ice may be used, but the Control Unit will not be able to control temperature above 50°F, and the fluid pump will pump at full speed during treatment.

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GR2.1 Temperature Control Block Diagram



GR 2.1 CONTROL PANEL

- **Actual Temperature:** Temperature of the water leaving the Control Unit
- **Target Temperature:** “Target” or “Set” temperature of water leaving the Control Unit based on user setting.

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Temperature Control for GR2.0 Control Units

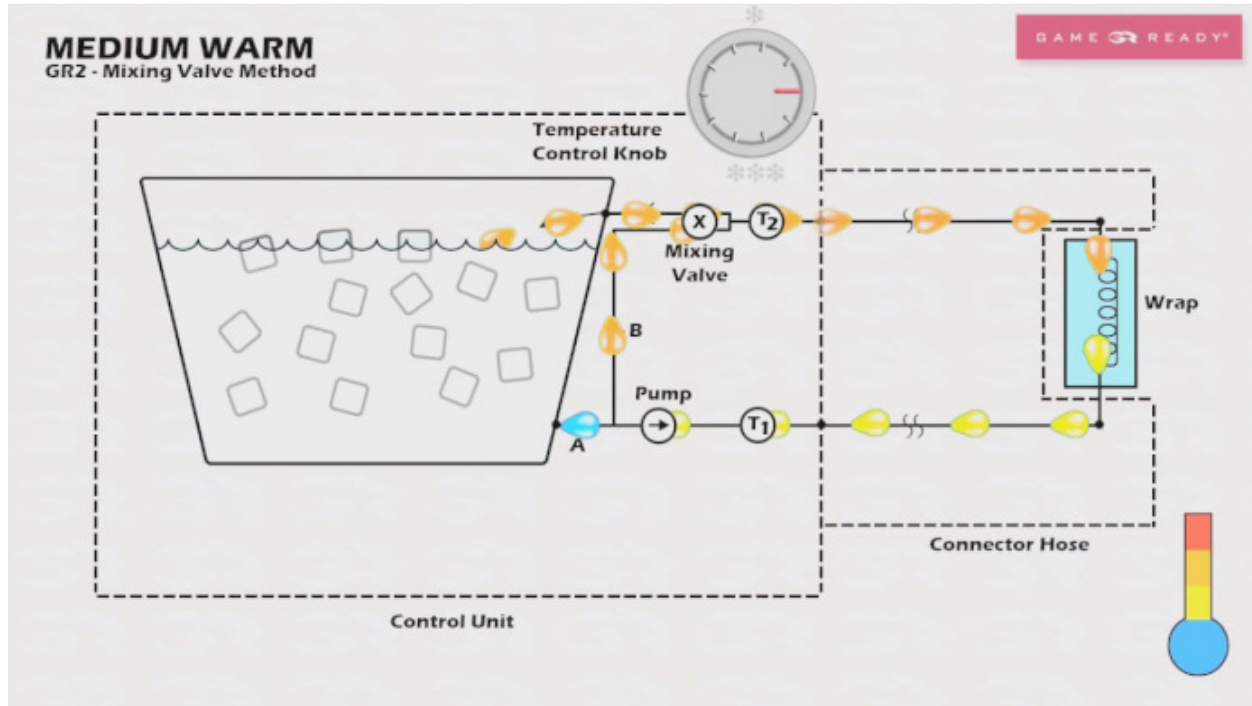
GR2.0 Control Units use a mixing valve connected to the temperature control knob. A ratio of cold water is mixed from the ice box and the warmer return water as shown below. At “Full Cold”, the pump is drawing 100% of its water from the ice box. At “Full Warm” the pump is drawing 100% of its water directly from the Wrap, recycling the return water which continues to be warmed by the body. At knob positions in between, the pump draws water from both the ice box and the Wrap, proportional to the position of the knob. Since the water temperature depends on the ice box temperature, return water temperature (heat load), and knob position, the temperature may drift over time, and the knob position will not always yield the same temperature each time. Temperature of the water is measured in two places in the GR 2.0 Control Units; referencing the figure below, the temperature is measured just before leaving the Control Unit (T1), and upon returning from the Wrap, back into the Control Unit (T2).

Two different temperatures are displayed on the Control Panel. The “Coldest Wrap Temp” is the temperature of the water leaving the Control Unit. The “Average Wrap Temp” is the average of the water leaving the Control Unit (T1) and temperature of the water returning to the Control Unit (T2).

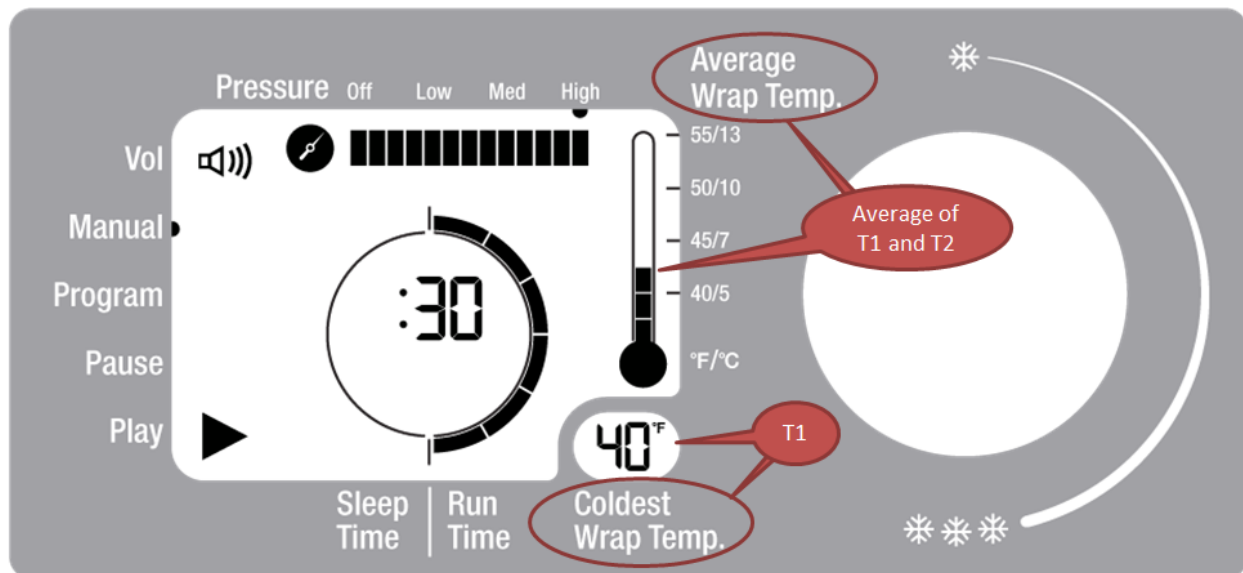
The following notes regarding the GR2.0 Control Unit also may be of interest:

- The fluid pump will always run at full speed, regardless of the temperature setting.
- If the ice box runs low on ice, there may not be enough ice to reach the desired temperature.
- When starting a treatment with a GR2.0 Control Unit, it is best to start with the knob at full cold. This ensures that water is being drawn from the ice box and reduces the chance of an air bubble entering the fluid pump and stopping fluid flow. If this happens, an “04 error” (Dry Pump Alarm) will occur. Reference the trouble shooting section of the manual for advice on clearing this error.
- There is no maximum temperature setting on a GR2.0. If set to full warm, the water will not draw water from the ice box, and will continue to recycle return water from the Wrap which is continuously warmed by the user’s body heat, and the water temperature in the Wrap could eventually reach body temperature.

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GR2.0 Temperature Control Block Diagram



GR 2.0 CONTROL PANEL

- **Average Wrap Temp:** Average temp of water leaving the control unit and returning to the Control Unit
- **Coldest Wrap Temp:** Temperature of water leaving Control Unit

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3. CLEANING

3.1. Control Unit

The exterior of the Control Unit and the visible interior surfaces of the ice box can be cleaned with warm water and mild soap, being careful to wipe with a damp cloth to remove any soap residue.

Caution: The Control Unit is not waterproof. Do not apply a direct stream of water onto or submerge the Control Unit. Do not allow fluid to pool on the surface of the face plate of the unit. Fluid left to pool on the face plate may accumulate under the surface and corrode the connections.

Disinfection may be performed using CaviWipes™ or the equivalent (CoolSystems has tested only CaviWipes). To ensure maximum effectiveness and compliance with applicable OSHA, CHAP or Joint Commission requirements, follow the manufacturer's recommendations for tuberculocidal activity. Cavicide also leaves a residue that, after it has dried, should be wiped off with a clean damp cloth.

CaviWipes may be purchased from numerous sources including the manufacturer's website at www.metrex.com.

Procedure:

- Follow the manufacturer's instructions and precautions for the cleaning agent
- Clean the Control Unit of visible dirt and debris
- To disinfect, wipe down all surfaces of the Control Unit so that the surfaces remain visibly wet for 3 minutes (at room temperature) or per manufacturer's instructions
- After 3 minutes, wipe the treated surfaces with a clean damp cloth to remove the disinfectant residue
- Allow the Control Unit to dry thoroughly before returning it to service or storing it in the bag
- If not returning the Control Unit to service right away, leave the lid to the ice box open to prevent mold/mildew formation
- The Control Unit should be disinfected whenever it is used on more than one patient, prior to being used by subsequent patients

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Do Not Use:

- Quaternary Ammonium (such as Virex)
- Phenolic-based disinfectants (such as Amphyl).
- Any solvent-based cleaners on the Control Unit. Doing so will damage the plastics and will void the warranty.
- Abrasive materials to clean the Control Unit. Doing so will damage the plastics and will void the warranty.
- Hydrogen Peroxide

3.2. Wraps (Heat Exchangers and Sleeves)

The Wraps are non-sterile products made up of outer Sleeves with a “hook and loop” closure (similar to Velcro®) and zippered pockets into which Heat Exchangers (the inner component through which the air and water flow) are inserted. Both may be cleaned of dirt and other biological soil for reuse or discarded, but neither can be sterilized.

Regular Maintenance/Cleaning

- Remove Heat Exchanger from Sleeve after each use to limit condensation collecting inside sleeve.
- The Heat Exchanger is made of fabric and, if it becomes soiled, it may be cleaned with Steri-Fab® (www.sterifab.com) per the manufacturer’s instructions or it may be discarded. The Heat Exchanger may also be cleaned with a soft cloth and warm soapy water followed by thorough rinsing and drying of the fabric.
- The Sleeve is intended for single patient use in the hospital setting and limited multi-patient use in outpatient settings with protected Sleeve-skin contact. Additional sleeves may be purchased by contacting Customer Service.
- For limited multi-patient use in outpatient settings, the Sleeve may be cleaned by spraying it down with Steri-Fab and/or washing on the gentle cycle in a standard washing machine. The Sleeve may be used and washed in this way for approximately 3 months or approximately 200 open/close cycles and 3-4 machine washings. After that the Sleeve may begin to lose pull strength on the hook & loop closure and should be replaced.

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Mildew

Condensation occurs naturally on the cold surface of the heat exchanger. Constant use and immediate storage of the Game Ready Wraps can cause mildew growth in some circumstances. The Sleeve portion of the Wrap is meant to be a protective barrier for the Heat Exchanger that can be removed and discarded or cleaned (as described above). The Sleeve also has a coating on the inside that locks in moisture. Like a wet umbrella, it needs to be allowed to dry or wiped down before storing. Failure to do so can result in mildew formation on the interior of the sleeve or on the Heat Exchanger.

If mildew does occur and the choice is to clean rather than discard:

- The Heat Exchanger may be cleaned carefully with Steri-Fab (noted above), which is also a mildewcide, following the manufacturer's instruction.
- The Sleeve may be washed, thoroughly dried and sprayed with Steri-Fab and again allowed to dry thoroughly before inserting a Heat Exchanger.

Mildew stains cannot be removed from the Sleeve or the Heat Exchanger. They can be cleaned, but the stain will remain. We recommend replacing the affected components and diligent preventative care to prevent future mildew growth. The following are some suggestions for avoiding mildew propagation.

- Drain most of the residual water from the Wrap following use by leaving it connected to the Control Unit for 5-10 minutes. This drain action can be enhanced by raising the height of the Wrap above the Control Unit as much as possible.
- Separate the Sleeve and the Heat Exchanger to dry them independently.
- Open the Sleeve and its zippered pocket as much as possible and dry off the interior blue surface. Leave open to thoroughly dry for as long as practical.
- Lay the Heat Exchanger flat and wipe dry and hang or lay flat on a towel to allow to dry.

3.3. Connector Hose

- The Connector Hose braided nylon cover may be cleaned with Steri-Fab and the plastic connectors may be cleaned with warm soapy water. The connectors may also be disinfected with CaviWipes.
- A PVC-covered Hose (plastic covering over the braided fabric for easier cleaning/disinfection) is available and may be ordered through Customer Service. (Game Ready Part # 570304)

TECHNICAL MANUAL**Do Not Use:**

- Quaternary Ammonium (such as Virex).
- Phenolic-based disinfectants (such as Amphyl).
- Any solvent based cleaners on the Connector. Doing so will damage the plastics and will void your warranty.
- Abrasive materials to clean the Connector. Doing so will damage the plastics and will void your warranty.
- Any petroleum based lubricants on the Connector. Doing so may damage the O-rings and will void your warranty.

3.4. Carry Bag

The Carry Bag should be cleaned using a soft cloth or brush and, if needed, decontaminated with Steri-Fab according to the manufacturer's instructions. Be sure to test any cleaner on a small portion of the Bag to make sure that it will not damage the Bag.

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4. MAINTENANCE / REPAIRS

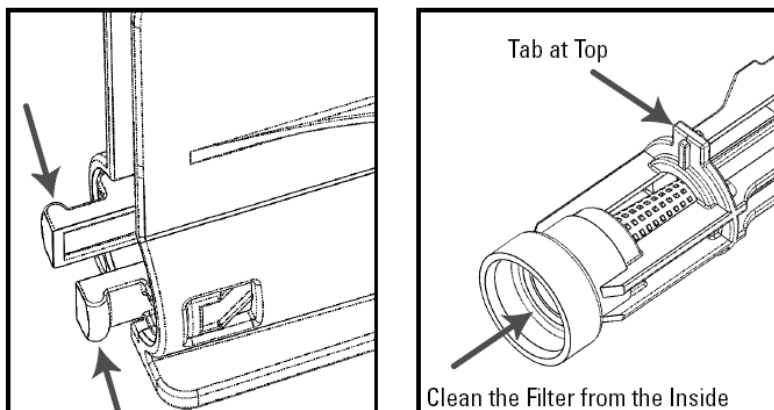
The following section contains certain maintenance and repairs that may be performed by users. While the maintenance and repair instructions are noted below, always be sure to use the instructions included with the replacement parts for additional information, cautions and warnings.

4.1. Filter

The Ice Box Filter helps protect against debris from damaging the fluid pump, and is found inside the Ice Box. If the filter becomes dirty or clogged, water flow thru the system will be restricted. It is recommended to inspect the ice box filter approximately once every 3 months or as needed:

Model 550550-X3 (GR2.1)

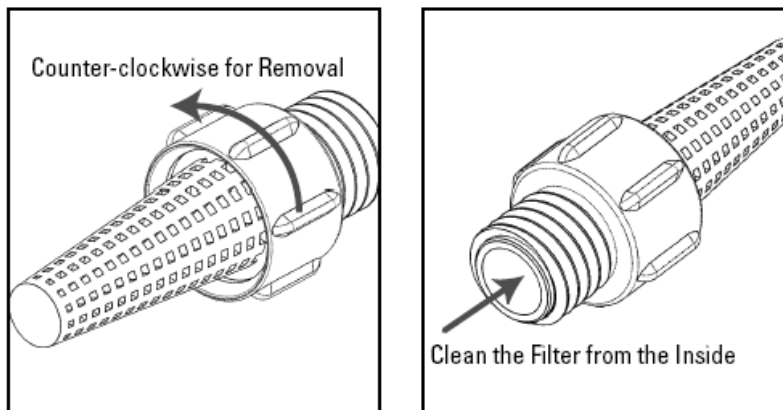
- Remove the filter by pressing the tabs together and pulling straight out.
- Clean the filter by rinsing it thoroughly under warm running water, or blowing with air thru filter mesh from the inside.
- Replace, orienting the Tab at the top, and pressing into place, ensuring tabs snap into place.



Model 550550-X2 (GR2.0)

- Remove the filter by rotating it counter-clockwise. If needed, a $\frac{3}{4}$ " deep socket wrench may be used.
- Clean the filter by rinsing it thoroughly under warm running water, or blowing with air thru filter mesh from the inside.
- Reinsert the filter "finger tight". Avoid cross-threading.

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Replacement filters may be obtained by contacting Game Ready Customer Service. Refer to the “Replacement Parts” section and order the filter associated with your model number.

4.2. O-rings

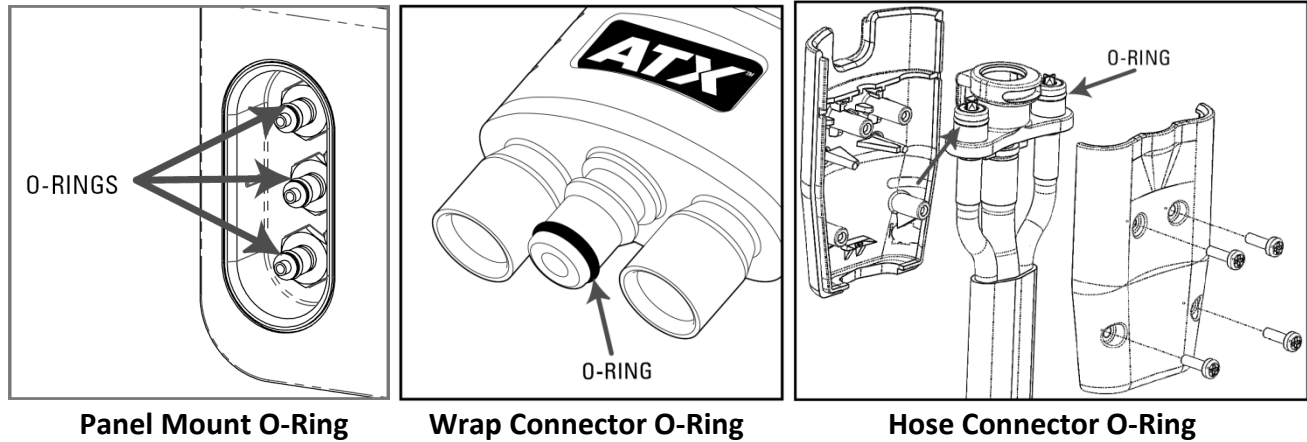
If the Hose or Wrap connections get “sticky” or hard to connect, they may require lubrication. Apply a light amount of silicone lubricant to the O-rings as needed. (In an emergency, water may help.) Note: To prevent possible damage to the O-rings, use only a silicone lubricant (i.e. not a petroleum based lubricant). This may be purchased by contacting your Game Ready Distributor or Game Ready Customer Service and ordering Silicone Pump Spray – (**Game Ready Part # 573201**).

The following alternate silicone sprays have also been successfully tested for use on the O-rings:

- DuPont Teflon Silicone Spray (DuPont Part # DS0614101)
McMaster Carr Part # 8710T22
www.mcmaster.com
 - 3M Silicone Spray (3M ID#62-4678-4930-3)
McMaster Carr Part # 31155K36
www.mcmaster.com
- or**
- 3M e-store
www.3mestore.com

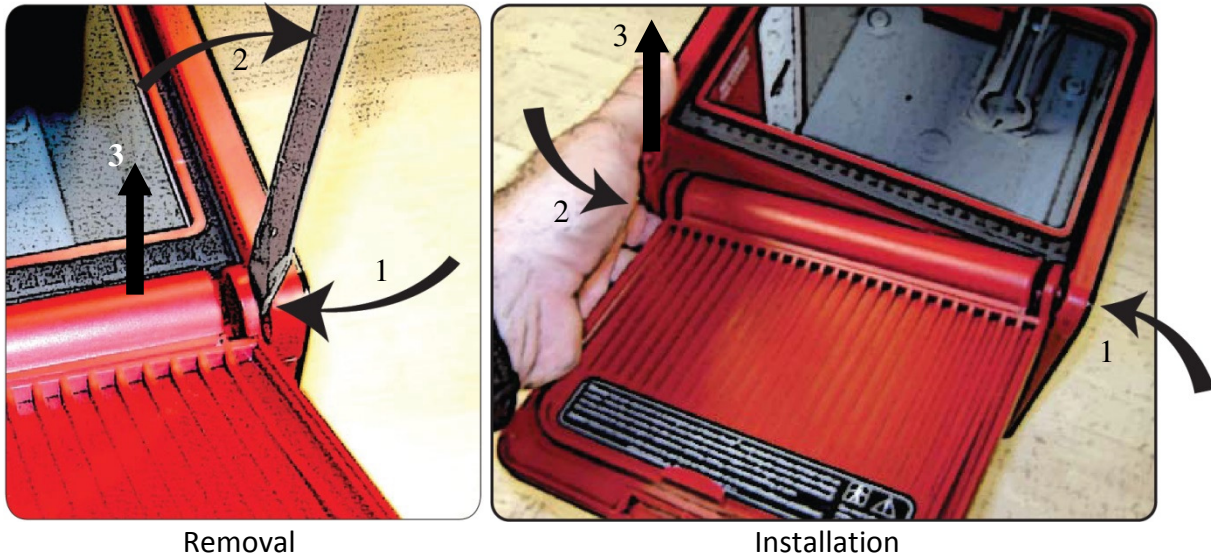
If the O-rings are cracked or damaged (i.e. leaking), they may be replaced using one of our O-ring replacement kits (Reference the *Replacement Parts* section):

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4.3. Door Lid

If the Ice Box door lid becomes damaged or broken, it may be replaced by ordering an Ice Box Lid Replacement Kit. Refer to the “Replacement Parts” section for the part number and description. Abbreviated instructions are shown below.



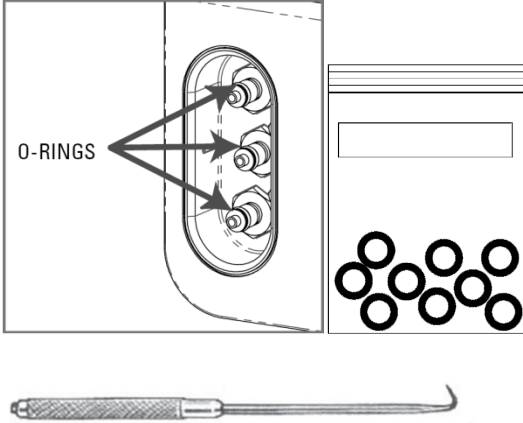
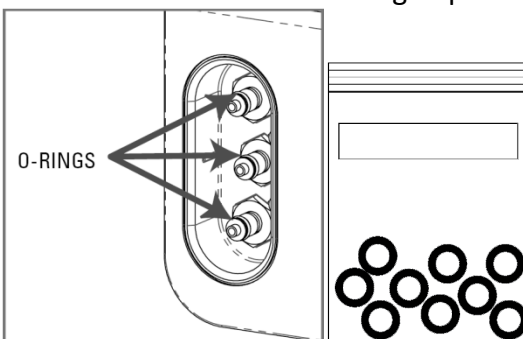
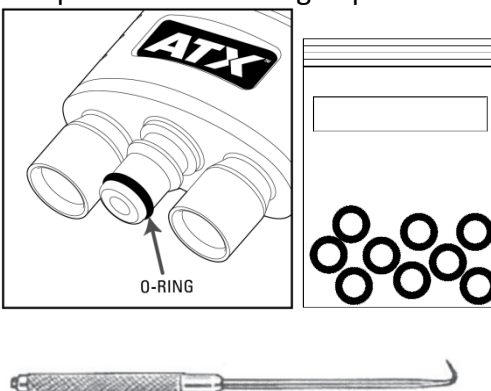
4.4. Other Repairs

Other than those listed above, no product repairs should be performed by the user. If your Game Ready System requires repair service, please contact your Game Ready Distributor or Game Ready Customer Service. Opening the Control Unit case will void the warranty.

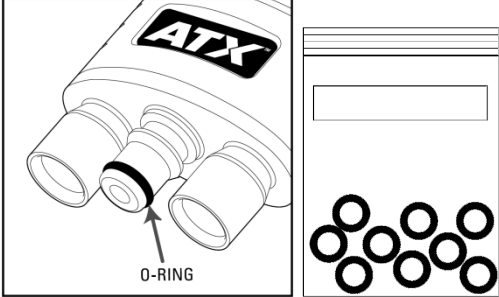
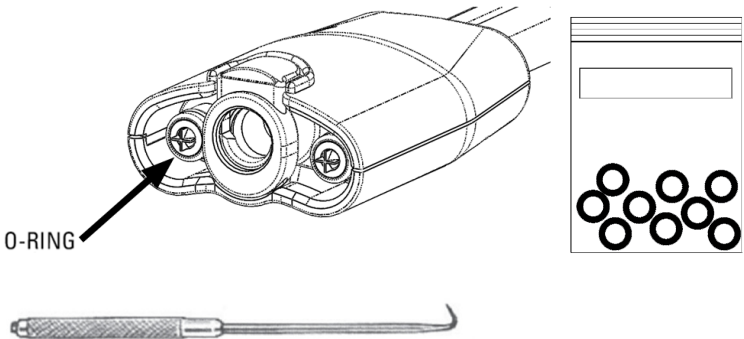
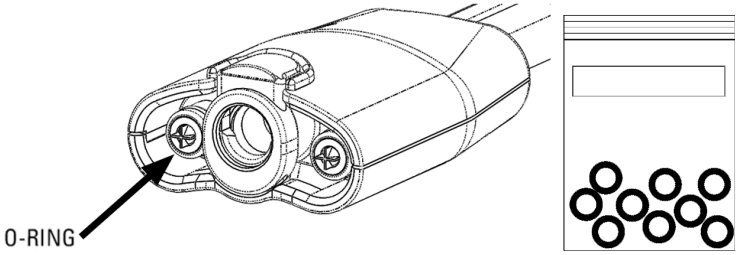
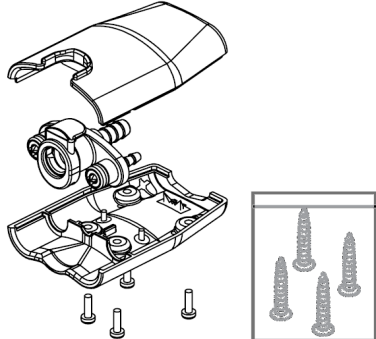
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4.5. Replacement Parts

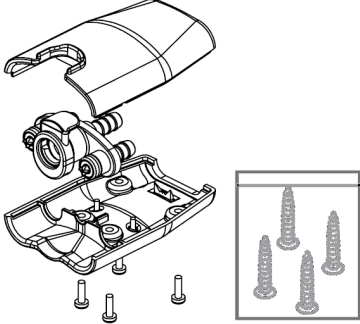
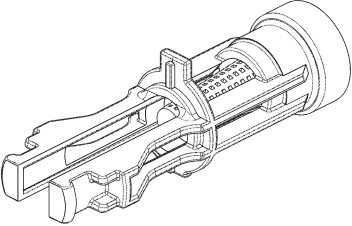
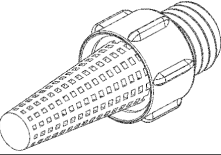
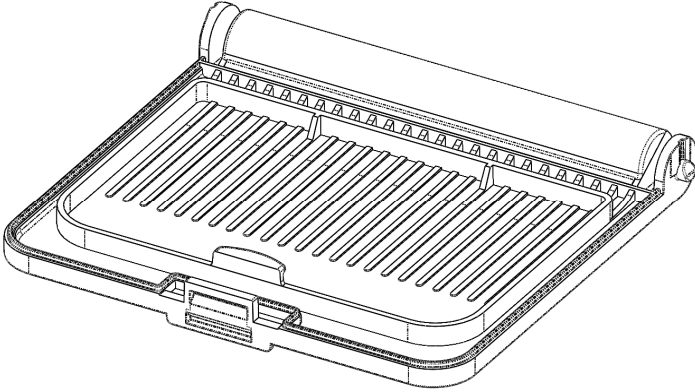
The following is a list and description of replacement parts and test fixtures that may be purchased thru Game Ready. Please contact your Game Ready Distributor or Game Ready Customer Service for more information.

Part #	Description	Instructions Part #
503512	Panel Mount Connector O-Ring Replacement Kit with Tool 	703793
503513	Panel Mount Connector O-Ring Replacement without Tool 	703793
573208	Wrap Connector O-Ring Replacement Kit with Tool 	704094

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573209	Wrap Connector O-Ring Replacements without Tool 	704094
573217	Hose Connector O-Ring Replacement Kit with Tool 	704134
573218	Hose Connector O-Ring Replacements without Tool 	704134
573214	3-Port Connector Replacement Kit (Blue Button) 	702448

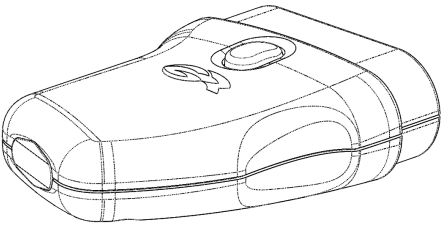
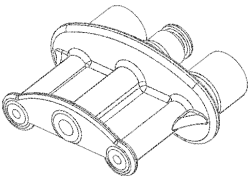
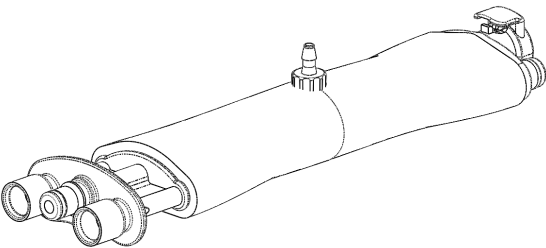
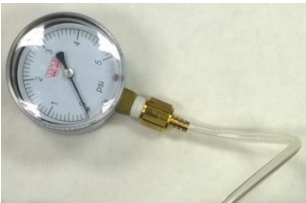
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503339	3-Port Connector Replacement Kit (Gray Button) 	702448
573216	GR 2.1 Replacement Filter 	704132
573215	GR 2.0 Replacement Filter 	704130
573219	Ice Box Lid Replacement Kit 	703913
573201	Silicon Pump Spray Package 	703512

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503510	AC Adapter with Detachable Cord Set, GR 2.1 and 2.0 Models	n/a
303528	AC Adapter, GR 2.1 and 2.0 Models	n/a
302506	Power Supply Detachable Cord Set, US	n/a
303022	Power Supply Detachable Cord Set, EU	n/a
303199	Battery Charger with US cord	n/a
303529	EU Battery Charger Power Cord	n/a

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570700	Unit Diagnostic Tool 	703440
502945	Drain Mode Connector 	n/a
502844	In-Line Pressure Test Tool 	703190
503511	Pressure gauge for In-Line Pressure Test Tool 	703190
573222	Temperature Verification Assembly 	704187

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5. CALIBRATION / VERIFICATION

The Game Ready System does not require calibration. The system components are verified for accuracy during the manufacturing process.

5.1. Pressure Verification

The accuracy of the air pressure generated by the system can be verified periodically using a test fixture available from Game Ready (see Figure 1). To obtain a test fixture, call your Game Ready Distributor or Game Ready Customer Service. To perform this test, the following equipment will be needed:

- The Inline Pressure Test Fixture (Part # 502844)
- The Control Unit to be tested
- Connector Hose
- AC Adapter
- Game Ready Straight Knee or Straight Elbow Wrap. (Other Wraps may also be used)
- A calibrated pressure gauge with a length of 1/8" ID tubing (i.e. Part # 503511)

Procedure:

- Set up the Control Unit
 - Fill the unit's reservoir with water (no ice is needed for this test).
 - Power the unit.
 - Attach the Connector Hose to the unit.
- Roll the Knee Wrap into a cylindrical shape approximately 3 to 4 inches in diameter to prevent it from ballooning.
- Connect the Inline Pressure Test Fixture between the Connector Hose and the Wrap. Attach the pressure gauge to the test fixture (See figure 1)

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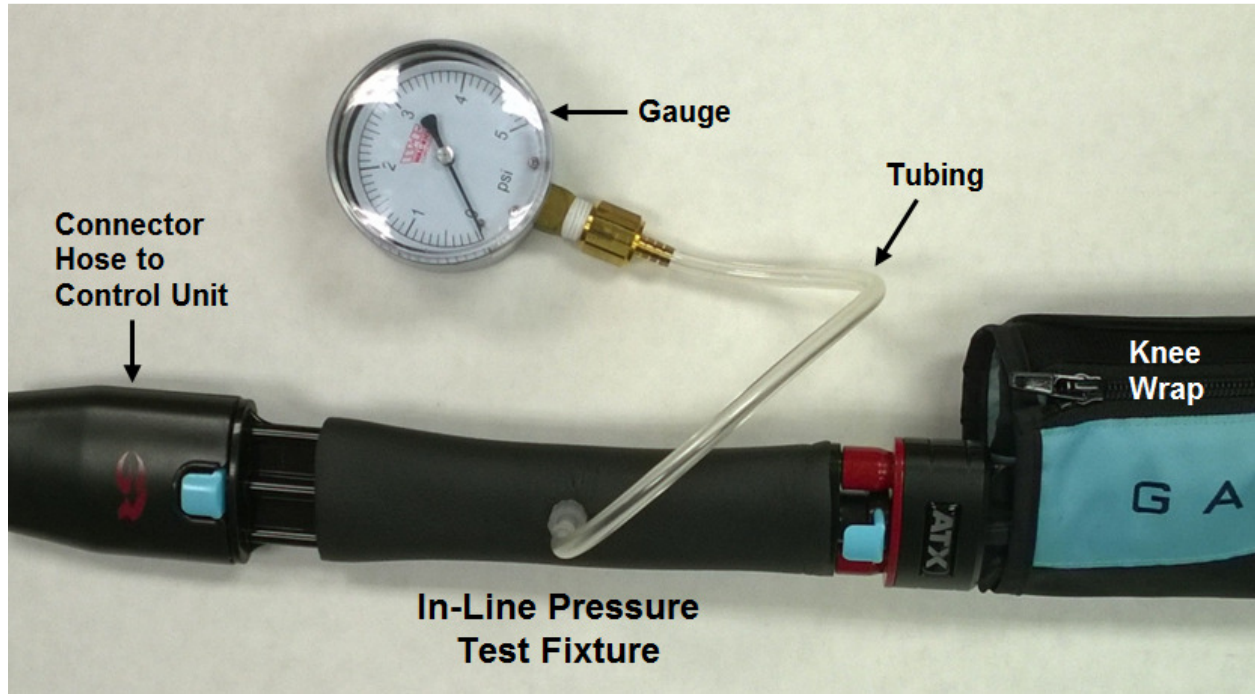
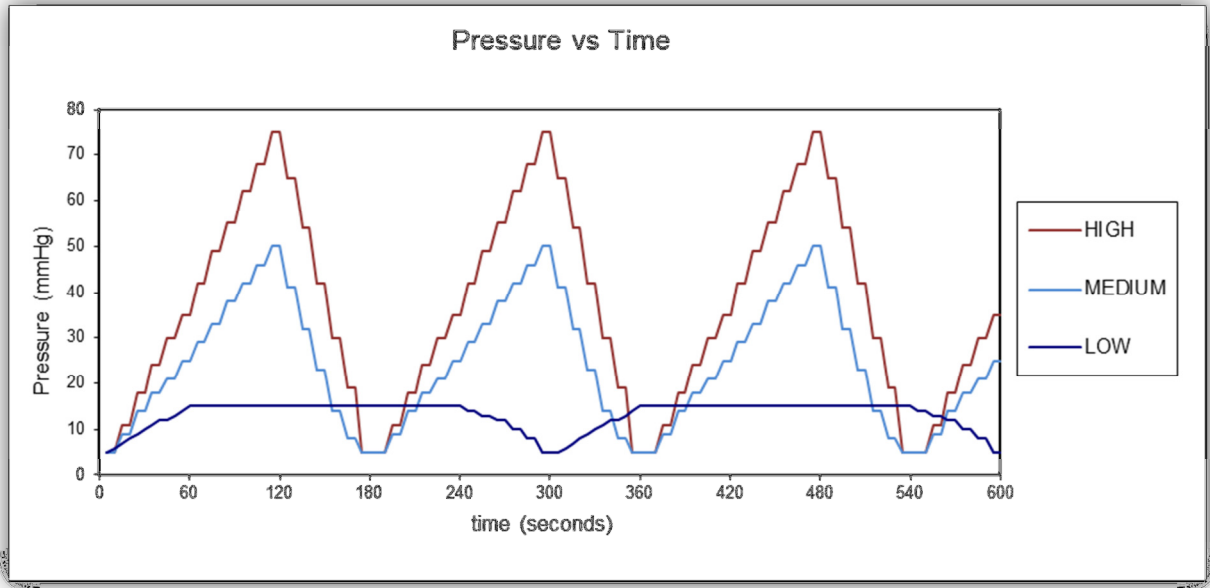


Figure 1

- Power the Control Unit on, and set it to the high pressure setting.
- Press the Control Unit's play button.
- As the Control Unit generates pressure inside the Wrap, that pressure will be displayed on the pressure gauge. The Control Unit will cycle between 0 and 75 mmHg (0-1.45 psi) at the high pressure setting over a 3-4 minute period.
- Observe the pressure gauge as the pressure increases. Approximately 2 minutes after pressing the play button on the Control Unit, the pressure will be at its maximum level.
 - Note: It may take longer than 2 minutes to reach the top of the pressure ramp, particularly on the first pressure cycle. You can tell the Control Unit has reached the top of the pressure cycle once the solenoid clicks open and the air pump stops cycling.
 - If you get a flow error (04 or 10) during testing, wrap the Wrap around a cylindrical object (like a 3-4" diameter mailing tube) to help prevent the fluid paths from kinking.
- Record the pressure displayed on the pressure gauge at the top of the pressure cycle. (Ignore quick pressure spikes when the pump turns on, and read the gauge after the pressure spike settles).

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- The pressure at the top of the High pressure cycle pressure should be 75 mmHg, +/- 17mmHg (1.45 +/- 0.33 psi). If the pressure is outside of this range, refer to the “Troubleshooting” section.

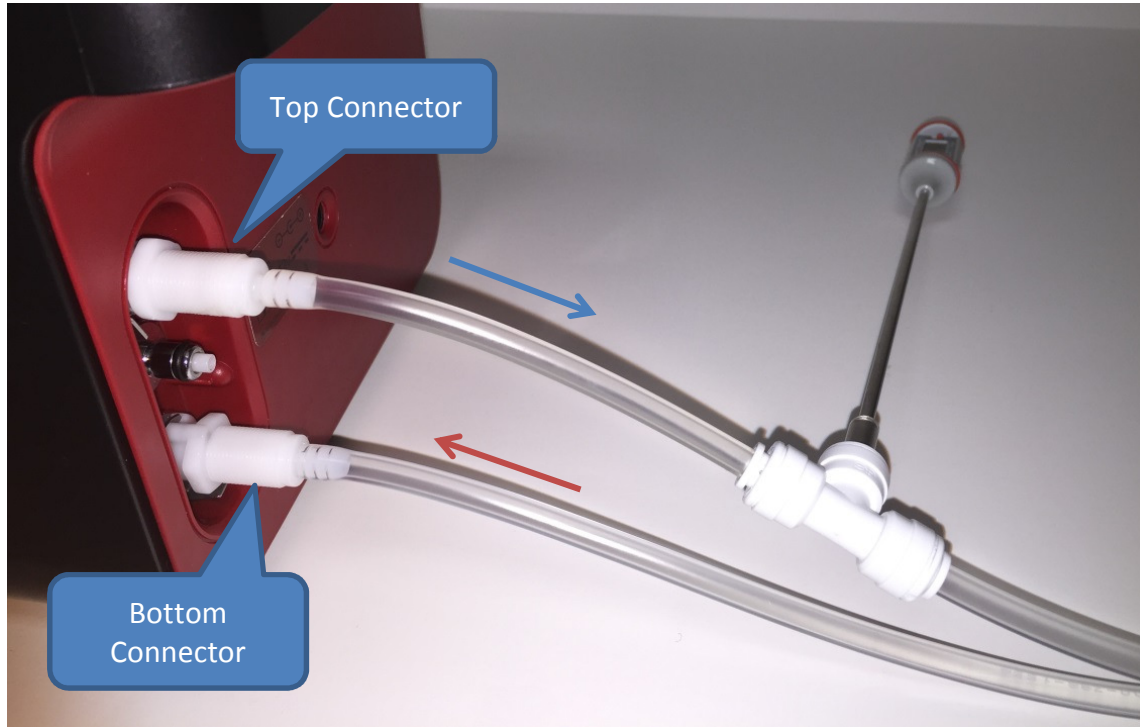


5.2. Temperature Verification

5.2.1. GR2.1 Temperature Verification

- Add water to the ice box to the fill line indicated inside the reservoir.
- Add ice to at least half full.
- Install the Temperature Test Fixture to the Control Unit per the figure below.
 - Note, make sure the Thermometer is connected to the Control Unit's **TOP** fluid connector and the connector returning from your Wrap is connected to the Control Unit's **BOTTOM** fluid connector.
- Connect a Wrap to the Temperature Test Fixture to use as a heat load during testing. Leave the Wrap open (flat) to provide sufficient heat.

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- Turn on the Control Unit and set the unit to “No Pressure” and the Target Temperature to 45-46°F.
- Press play. You will hear the pump flow at full speed until water circulates thru the Wrap and returns to the Control Unit. You will then

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hear the pump slow down as it begins to deliver water at the set temperature.

- Wait up to 5 minutes for the temperatures to stabilize.
- You should be able to confirm that the temperature on the Reference Thermometer is similar to the Target Temperature. You should also see correlation between the “Actual” Temperature Readout on the Control Unit and the Target Temperature.



- Expect some difference between the Target Temperature, the Reference Thermometer and/or the “Actual” Temperature Readout on the Control Unit are normal due to a number of factors including error between the two temperature sensors, dead band / hysteresis in the Temperature Display, difference in temperature measurement location, ambient temperature, Wrap size used, amount of ice and water in the Ice Box, etc. These compounding factors may lead to errors as high as +/- 4°F or more. Minor temperature fluctuations /

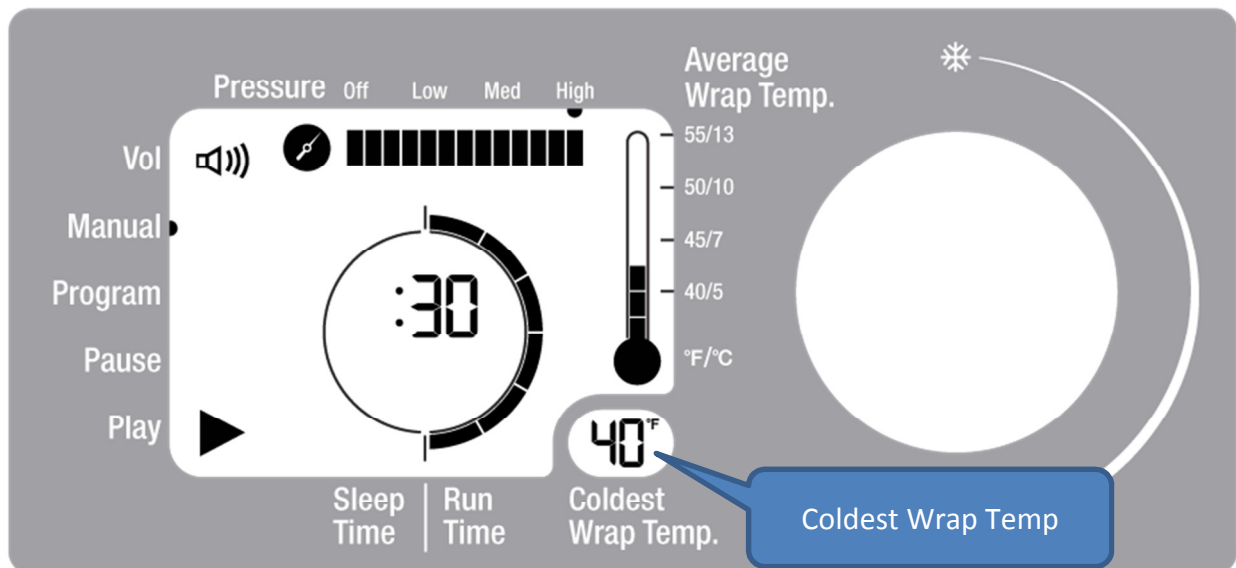
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oscillations in temperature are also to be expected. In general, you should find that the Control Unit adjusts temperature adequately, and this test should verify that the temperature control system is functional.

- If the Reference Thermometer consistently reads more than 5 °F colder than the Target Temperature, consult the Troubleshooting section, or contact your distributor or Game Ready customer service.

5.2.2. GR2.0 Temperature Verification

- Fill the ice box at least half full with ice and water.
- Install the Temperature Test Fixture and a Wrap per the previous section.
- Turn on the Control Unit and set the unit to “No Pressure” and turn the temperature adjustment knob on the Control Unit to “Full Cold”.
- Press play. The pump will run at full speed, drawing all of its water from the reservoir. Wait up to 5 minutes for the temperatures to stabilize.
- You should be able to confirm that the temperature of the reference thermometer is similar to the “Coldest Wrap Temp” shown on the Control Unit.



- Some difference between the “Coldest Wrap Temp” and the Reference Thermometer are normal due to a number of factors including error between the two sensors, dead band / hysteresis in the Temperature Display, difference in temperature measurement location, ambient temperature, Wrap size used, amount of ice and

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water in the Ice Box, etc. These compounding factors may lead to errors as high as +/- 4°F.

- If the Reference Thermometer consistently reads more than 5 °F colder than the displayed “Coldest Wrap Temperature”, consult the Troubleshooting section, or contact your distributor or Game Ready customer service.

5.3. Button Functionality

The Control Panel may be functionally checked by pressing each button and observing proper response from the system. If any of the buttons are difficult to press, the Control Unit may be sent in for service.

5.4. Encoder Check (GR2.1 only)

The Temperature Adjustment encoder may be checked for functionality.

- Set the temperature display to °F.
- Slowly sweep the temperature knob from one position to the other.
- When set to display °F, each position should register a different “Target Temperature” between 34-50.
- There are two types of encoders. Both versions are shown in the tables below. Note that when displaying in °C there will be some redundant or missing numbers due to rounding error.

New Encoder		
Knob Position	°F	°C
0 (Full Warm)	50	10
1	48	9
2	46	8
3	44	7
4	42	6
5	40	4
6	38	3
7	36	2
8 (Full Cold)	34	1

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Older Encoder		
Knob Position	°F	°C
0 (Full Warm)	50	10
1	49	9
2	48	9
3	47	8
4	46	8
5	45	7
6	44	7
7	43	6
8	42	6
9	41	5
10	40	4
11	39	4
12	38	3
13	37	3
14	36	2
15	35	2
16 (Full Cold)	34	1

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6. STORAGE

When you are done using the System for the day:

- Unplug the AC Adapter and the Connector Hose from the Control Unit.
- Push the door release button to open the door.
- Carefully dump the ice and water into the sink or you may use the Drain Mode and the Drain Mode Connector to drain the water (see “Drain Mode” section below)
- Wipe off any excess water.
- The Control Unit may be stored inside or outside the Carry Bag.
- Store the Control Unit with the lid open to allow the interior to dry and preserve the ice box seal.
- Storage Temperature Conditions: 1° - 50° C (33° - 122° F)
- Relative Humidity: 10% - 95% non-condensing
- Do not store the AC Adapter inside the ice box, as there may be some residual moisture which may eventually damage the AC Adapter.

Caution: Storing your system in extreme temperatures (above 122°F or below 33°F) may cause damage. Do not leave in a hot or freezing car. Do not leave the Control Unit in direct sunlight.

Drain Mode:

This mode allows a user to connect a Hose with a special Hose Adapter (Drain Mode Connector) to the unit, allowing one to pump the water out of the ice box to reduce the weight of the Control unit before lifting. A Drain Mode Connector may be purchased by contacting Customer Service.

Drain Mode can be accessed by pressing the Program key and toggling through all six programs until the Drain Mode icon appears:

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To empty water from the system while in Drain Mode, attach the Drain Mode Connector to the Connector Hose, place the Hose over a sink, and press the play button. The Control Unit's fluid pump will run for six minutes (long enough to fully empty the ice box). Once the pump stops, the Control Unit will beep once and return to Manual Mode. Unplug the Control Unit and dump any remaining ice and water into the sink.

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7. PRODUCT SPECIFICATIONS

Model # 550450-XX and 550550-XX
Size: 16.25" length x 7.75" width x 9.25" height (413 x 197 x 235) mm, not including carrying case
Weight: 7.3 lb. empty, approximately 18 lb. full of ice and water
Pressure level: cycles from 5mm Hg up to 75mm Hg
AC power: 100-240 V~, 50-60 Hz, 1.6A
Control Unit Power input: 12V DC/2.5 A

7.1. SAFETY INFORMATION

Model # 550550-XX	
	Medical Equipment With respect to electrical shock, fire, and mechanical hazards only in accordance with UL 60601-1, CAN/CSA C22.2 No. 601.1
 12V/2.5A	Powered by Direct Current
	Type BF Applied Parts

7.2. UL CLASSIFICATION

Protection against electric shock:

The Game Ready System is considered to be Class I (Protective Earth) when connected to the FSP Model # FSP030-RCAM power supply. The Game Ready System is considered to be Class II (double insulated) when connected to the GlobTek Model GTM-21097-5012 power supply.

Protection against harmful ingress of water:

This product provides ordinary protection against ingress of water.

TECHNICAL MANUAL**Degree of safety in the presence of flammable anesthetics or oxygen:**

Not suitable for use in an oxygen enriched environment or in the presence of flammable anesthetics.

Pollution degree classification:

This product is classified as pollution degree 2.

Electromagnetic interference:

This equipment has been tested and found to comply with the limits for medical devices in IEC 60601-1-2:2001. These limits are designed to provide reasonable protection against harmful interference in a typical medical installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to other devices in the vicinity. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to other devices, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving device.
- Increase the separation between the equipment.
- Connect the equipment into an outlet on a circuit different from that to which the other device(s) are connected.
- Consult the manufacturer or field service technician for help.

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8. TROUBLESHOOTING

Error / Problem	Possible Causes	Action
Air Pressure Sensor  This error occurs if the system senses non-zero pressure during startup. This error can only occur at startup.	This may occur if a partially deflated Wrap is attached to the system during power up.	Disconnect Hose from Wrap and power up the Control Unit.
	Air pressure sensor is malfunctioning	With the Hose disconnected from the Wrap, attempt to re-start the unit a few more times, or a few hours later. If the 01 error does not resolve, the unit will require service.
Self Test - Air Pump:  This error means that the air pressure sensor did not sense the air pump turning on during start up. This error occurs during the self-diagnostics and therefore can only occur at startup.	Air pump is weak or nonfunctional and unable to turn.	Try re-starting the Control Unit several times. Attach a Hose to the Control Unit to ensure the air pump can start more easily. If the 02 error continues to occur, the unit will require service.
	Air pressure sensor not working or disconnected.	The unit will require service.
Dry Pump  This error is comparable to the Low Flow (10) Error except this should only happen at the beginning of a treatment. This error occurs when “play” (the start/pause button) is pushed and return flow is not established within the first 45-60 seconds.	Not enough water in Control Unit (i.e. runs low on water when filling a Wrap for the first time)	Make sure there is adequate water in the Ice Box. 2-3 inches is all that’s necessary.
	“Dry” pump – the fluid pump may have an air bubble in it and may need priming.	With a GR2.0 Control Unit, it is important to start the treatment <u>at full cold</u> to reduce the chance of an air bubble entering the pump. For either system, the pump may be primed as follows: • Disconnect Hose from Control Unit. • Turn on Control Unit. • Using a towel or edge of your fingernail (or otherwise <u>not blocking the outlet</u>) press the <u>top white</u> connector valve and press Play. Water should squirt from the valve, indicating the pump is now primed.
	Wrap takes too long to fill for the first time, and causes an error. This is more likely if using a dual Hose since it takes twice as long to fill the Wraps.	To Prime the Wrap: 1. Place the Wrap on a flat surface such as a table or the floor. 2. Ensure the Connector Hoses are fully connected by hearing the ends snap into place 3. Set the system to run with no pressure • Set the temperature knob to full cold

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Error / Problem	Possible Causes	Action
		(GR2.0 only) Press play (the start button) and allow to run for at least a minute.
	Clogged filter	Clean or replace the Ice Box filter if dirty.
	Wrap folded or kinked during application	Disconnect the Hose from the Wrap to allow the air to vent, lay Wrap flat on a table and run with no pressure for at least 1 minute. Reapply Wrap.
	Wraps applied too loosely may develop kinks when using higher air pressure settings.	Disconnect the Hose from the Wrap to allow the air to vent and re-apply the Wrap more snugly. If this doesn't work, try using the system at a lower pressure setting.
	Hose kinked	Check that the Hose is not kinked, and that the Wrap is not being pulled near the Hose connection.
	Hose or Wrap disconnected.	Check Hose connection at the Control Unit and Wrap. Reconnect, listening for a "click".
	Flow Sensor malfunction (false alarm)	Try using other Wraps to rule out Wrap kinks or application errors. If you continue to get errors, or if you are able to prime the Wrap and visually see strong return water flow into the ice box and you still get an 04 error, there may be a malfunction in the flow sensor. In this case, the Control Unit will require service.
Over Pressure Error  This Error indicates that the unit has exceeded the target pressure set by the user.	Excessive movement by the patient during treatment.	Reduce movement during use.
	Error with the pressure sensor or air pump control.	Try using at a lower pressure. If the 06 error persists the Control Unit will require service.
Under Pressure Error  This Error indicates that the unit failed to reach its target pressure within a specified period of time	This can occur if the Wrap "opens up" during use due to fatigued hook and loop ("Velcro") on the sleeve. This will cause the air bladder to "balloon" and become too large to fill in the allowed time and an error will be called.	If the Velcro is failing during use, replace the sleeve. Check to make sure the zipper on the sleeve is closed.
	If the Wrap is applied too loosely, (especially using a dual Hose) the air bladder may "balloon" and become too large to fill in the allowed time.	Disconnect the Hose from the Wrap to allow the air to vent and re-apply the Wrap more snugly.
	Using a dual Hose with only one Wrap attached.	If using a dual Hose, confirm that a Wrap is plugged in to <u>both Hoses</u> , as air will leak out of the unused Hose if only one Hose is connected to a Wrap.
	Leaky O-ring, either at the Control Unit (center connector) or the center Wrap Connector.	Listen for audible leaks, and try manipulating the Connector. The O-rings can be replaced using an O-ring replacement kit (Refer to "Maintenance/Repair" Section).
	Leak in the air bladder of the Wrap or	Try using a different Wrap or Hose to isolate

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Error / Problem	Possible Causes	Action
	in the Hose	the problem.
	Air leak inside the Control Unit	If all of the above does not resolve the issue, there may be an air leak inside the Control Unit and it will require service.
Deflation Error  This error occurs when the Wrap fails to deflate as expected. After reaching the maximum pressure, the unit has two minutes to deflate to 15 mmHg air pressure.	This error could occur if the “vent” in the Control Unit does not open or is clogged, or if there is an error in the pressure sensor circuitry.	This error is highly unlikely to occur, however if a Control Unit has this error, it will need to be serviced.
Pump Performance Error  This error occurs when the fluid pump is drawing excessive current.	The fluid pump may be working too hard.	• Check the Ice Box Filter and clean or replace if dirty. • Ensure there are no kinks in the Wrap (See 04 and 10 errors) • Check Hose connections • Allow the pump to cool off for 20 minutes and re-try.
	The fluid pump may need replacement.	The Control Unit will require service.
Low Flow Error  This error is similar to an 04 error, except it occurs only after return flow has already been established. Depending on your model, a Low flow error will require a reduction in flow for at least 60 seconds. Note, for GR2.1 systems, when set to warmer temperatures, flow will be low to begin with and the flow sensor will not be monitored. If the outlet water temperature increases too far above set point, voltage to the fluid pump will increase until maximum pump speed is reached. At	Not enough water in Control Unit (i.e. runs low on water when filling a Wrap for the first time)	Make sure there is adequate water in the Ice Box. 2-3 inches is all that’s necessary.
	Wraps applied too loosely may develop kinks when using higher air pressure settings.	Disconnect the Hose from the Wrap to allow the air to vent and re-apply the Wrap more snugly. If this doesn’t work, try using the system at a lower pressure setting.
	Wrap folded or kinked during application	Disconnect the Hose from the Wrap to allow the air to vent, lay Wrap flat on a table and run with no pressure for at least 1 minute. Reapply Wrap.
	Hose kinked	Check that the Hose is not kinked, and that the Wrap is not being pulled near the Hose connection.
	Hose or Wrap disconnected.	Check Hose connection at the Control Unit and Wrap. Reconnect, listening for a “click”.
	Clogged filter	Clean or replace the Ice Box filter if dirty.
	Flow Sensor malfunction (false alarm)	Try using other Wraps to rule out Wrap kinks or application errors. If you continue to get errors, and if you are able to prime the Wrap and visually see strong return water flow into the ice box and you still get an 04 error, there may be a malfunction in the flow sensor. In this case, the Control Unit will require service.

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Error / Problem	Possible Causes	Action
this point, the flow sensor will be monitored for a low flow condition.		
Self Test - Fluid Circuit:  This error means that the Fluid pump is not drawing any current, indicating a problem with the pump motor. This error occurs during the self-diagnostics and therefore can only occur at startup.	The continuity between the motor brushes may be interrupted.	Try re-starting the Control Unit several times. If the problem persists, the Control Unit will require service.
Temperature Sensor Error  The Control Unit has detected a temperature beyond expected limits. This error occurs during the self-diagnostics and therefore can only occur at startup.	The Control Unit was stored in extreme cold or heat	Do not store the system in extreme temperatures, as you may damage your system. Allow the system to reach room temperature and re-start the system. If the unit was stored in a hot location, the cool down process may be accelerated by filling the unit with ice, hook up to a Wrap, and re-start the unit several times.
	The Temperature Sensor is malfunctioning	The Control Unit will require service.
Control Unit won't power up (No Orange Light on Control Panel when plugged in)	AC Adapter not providing power	• Check the connection between the AC Adapter and the Power Cord. • Make sure the AC Adapter is plugged into a working electrical outlet and all connections are secure. (Test outlet with another electrical device. • Check for a Green Light on the AC Adapter. If the green light on the Adapter is off, then the AC Adapter is likely bad. • Check for an intermittent connection at the barrel plug or at the AC Adapter by wiggling the strain relief while plugged into the Control Unit. If the Control Unit intermittently begins to receive power, the AC Adapter may need replacement. • Measure the voltage on the barrel plug (should be 12VCD +/- 10%) • Try a different AC adapter • If AC Adapter is not functional, contact Customer Service.
	AC Adapter is good, but the Control	Does the Green light on the <u>AC Adapter</u> flicker

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Error / Problem	Possible Causes	Action
	Unit does not show an Orange or Green Light, there may be a problem with the Control Unit.	when plugged into the Control Unit? If so, this likely means there is a short inside the Control Unit and it may have been caused by an AC Adapter supplying too much voltage. Contact Customer Service. They will likely ask that you send back the Control Unit and the AC adapter. Check your AC Adapter(s) for proper voltage.
Control Unit won't power up (Control Unit shows an Orange light, but will not power on)	It may be a problem with the keypad.	Try pressing the On button harder. If this solves the problem, the keypad may need replacement.
	Boot up issue	Try unplugging the barrel plug and plugging it back in <u>immediately</u> and try to power the unit up again.
Control Unit shuts down	When the Control Unit shuts down prematurely follow the steps below to troubleshoot:	If there is an Error message, check the appropriate troubleshooting action for the error displayed.
	Problem with electrical connections	Check all power cords and connections. Make sure the unit is not plugged into a switched power outlet.
	Problem with the AC Adapter	Follow the trouble shooting steps under "Control Unit won't power up"
Hose difficult to connect	Dry or swollen O-rings	Reference O-Ring replacement in the Maintenance / Repair section.
Wrap over-inflating / ballooning	Hook and Loop is opening up on the Wrap	If the Hook and Loop is opening, the sleeve likely needs replacement. Try using on a lower pressure setting
	Wrap is applied too loosely	Disconnect the Hose from the Wrap to allow the air to vent and re-apply the Wrap more snugly.
Leaking Control Unit	Condensation may form on the unit during the normal course of use (particularly when it is warm and humid out – like a cold can of soda). This can also happen on the internal components – therefore, it may appear as a "slow leak" from the bottom seam.	If condensation is causing a small amount of water to collect, place a towel under the unit or use in the Carry Bag to help insulate the unit. Continue to monitor for water to be sure it does not collect in the carry bag.
	Sometimes water collects on the ice box lid and gasket which may eventually drip down the back, which may make it appear that the unit is leaking.	Check the back of the Control Unit for water collecting under the back of the door.
	Leaking from Hose connection	Check for water leaking under the Connector. If the Connector is leaking, try a different

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Error / Problem	Possible Causes	Action
		Hose, or try replacing the O-rings on the Control Unit. Reference O-Ring replacement in the Maintenance / Repair section.
Wrap “not getting cold”	Not enough ice in the Ice Box	Make sure the Unit has both sufficient ice and water in the system.
	Temperature set too warm	Check the temperature control knob for the proper setting. 3 snowflakes = coldest setting.
	Wrap is partially kinked - If flow is limited, the water will not flow quick enough to provide even cooling thru the Wrap.	Disconnect the Hose from the Wrap to allow the air to vent and re-apply the Wrap more snugly and ensure there are no kinks in the Wrap. If this doesn’t work, try using on a lower pressure setting. You can monitor return flow into the ice box to estimate the flow rate. When set to Full cold, if you notice a difference in flow rate between a Wrap lying flat on the table, and applied to the user, you may need to adjust the Wrap per above.
Only one Wrap getting cold when using a dual Hose	If there is a kink in one Wrap when using a dual Hose, most or all of the flow may go thru the other Wrap.	Disconnect the Hose from the kinked Wrap to allow the air to vent and re-apply the Wrap more snugly and ensure there are no kinks in the Wrap. If this doesn’t work, try using on a lower pressure setting.
The lid won’t open	Sometimes pumping water from the ice box into a Wrap can cause a slight vacuum which makes it difficult to open the Control Unit lid.	Press the door release button and pull up firmly, and the door should release. If needed, stop treatment and allow water to drain back into the Control Unit.
Lid won’t stay closed	Door latch may be broken	Replace the door – reference the Door Replacement procedure in the Maintenance / Repair section.
The unit fails the “Pressure Verification” test described in the “Calibration / Verification” section.	Pressure is set to the wrong setting	Double check the pressure setting on the Control unit.
	Not monitoring pressure at the proper time in the pressure cycle.	It may take longer than 2 minutes to reach the top of the pressure ramp, particularly on the first pressure cycle. You can tell the Control Unit has reached the top of the pressure cycle once the solenoid clicks open.
	Pressure is “spikey”	There may be momentary pressure spikes when the air pump turns on. These are typically transient, and not felt by the user and are considered normal. Read the pressure gauge once the pressure spike settles.
	Pressure sensor needs re-zeroing	Disconnect the Hose from the Control Unit or the Wrap from the Hose and restart the Control Unit. This will check for a zero offset error and re-zero the pressure sensor if the

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Error / Problem	Possible Causes	Action
		zero offset is within tolerance. (If it is not within tolerance, the Control Unit will give an 01 error.) If you did not receive an 01 error, re-test the Control Unit.
The unit fails the “Temperature Verification” test described in the “Calibration / Verification” section: If the <i>Reference Thermometer</i> consistently reads more than 5 °F colder than the <i>Target Temperature</i> :	Mismatched units	Check that the Control Unit and the Reference Thermometer are both displaying °F
	Insufficient heat load	Try using a larger Wrap for the verification, or place a warm object on the Wrap.
	Excessively high water level in the ice box.	If the water level is too high, or the heat load too small, the water temperature in the baffles will not heat up quickly. Fill the Control Unit according to the fill level indicated inside the ice box and/or add additional heat load during the test.
	Inaccurate Reference Thermometer	Check the accuracy of the reference thermometer by pushing down on the retaining color on the Tee fitting while pulling out the thermometer probe. Soak thermometer probe in ice water while stirring. Temperature probe should be at 32 °F
	Thermistor accuracy is out of specification	Your Control Unit may require service.

TECHNICAL MANUAL**9. CONTACT US**

Contact your Game Ready distributor or Game Ready Customer Service at 1.888.426.3732 (from outside the US call +1.510.868.2100).

Before any Game Ready System or component can be returned for service, you must obtain a Returned Material Authorization (RMA) number. Game Ready Customer Service can provide that number or your Game Ready Distributor can assist with that process.

Additional information such as tutorial videos and the most up to date documentation on your Control Units and Wraps may be found here:

<http://www.gameready.com/support>